

Extended or Long Form of the Periodic Table

s-Block Elements		p-Block Elements																0		(18)														
Period ↓ ↑	IA (1)																																	
	IIA (2)																																	
	IIIA (13)		IVA (14)		VA (15)		VIA (16)		VIIA (17)																									
	IIIB (3)		IVB (4)		VB (5)		VIB (6)		VIIB (7)		VIII (8)		IX (9)		X (10)		XI (11)		XII (12)															
	IIIB (3)		IVB (4)		VB (5)		VIB (6)		VIIB (7)		VIII (8)		IX (9)		X (10)		XI (11)		XII (12)															
	IIIB (3)		IVB (4)		VB (5)		VIB (6)		VIIB (7)		VIII (8)		IX (9)		X (10)		XI (11)		XII (12)															
	IIIB (3)		IVB (4)		VB (5)		VIB (6)		VIIB (7)		VIII (8)		IX (9)		X (10)		XI (11)		XII (12)															
1	H 1.0079 Hydrogen																																	
2	Li 6.940 Lithium																																	
3	Na 22.990 Sodium																																	
4	K 39.098 Potassium																																	
5	Rb 85.468 Rubidium																																	
6	Cs 132.91 Cesium																																	
7	Fr 223 Francium																																	
*Lanthanide Series																																		
**Actinide Series																																		

Periodic Classification-TPA-1**Only one correct :**

1. In lanthanum ($Z = 57$), the 57th electron enters in a
(A) 6p orbital (B) 5d orbital (C) 6s orbital (D) 4f orbital
2. In Mendeleev periodic table total no. of groups are :
(A) 8 (B) 9 (C) 18 (D) 12
3. Atomic number of the inert gas of 7th period = 118. Which is correct IUPAC name of last element halogen family in 7th period -
(A) Ununoctium (B) Ununnilium (C) Ununennium (D) Ununseptium
4. Three elements X, Y, Z are following Dobereiner's Triad rule. If the atomic weight of X and Y are 10 and 26 respectively, then atomic weight of Z will be
(A) 34 (B) 40 (C) 42 (D) 18
5. The valence shell of the element X contains 1 electron in 5s subshell and below that shell, 4 electrons in 4d subshell. The element belongs to which group (IUPAC) in periodic table-
(A) 4th group (B) 5th group (C) 6th group (D) 7th group

More than one correct

6. An element has electronic configuration as $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^7$.
Correct statement regarding element is -
(A) Element must be Co
(B) Element belongs to group no. '9' according to the long form of periodic table
(C) Maximum no. of electrons in element having $m = +1$ are 6
(D) Element has magnetic moment 15 B.M.

Match the column :

7. Match List-I (atomic number of elements) with List-II (position of element in periodic table)

List-I

- (A) 19
(B) 22
(C) 32
(D) 64

List-II

- (P) p-block
(Q) f-block
(R) d-block
(S) s-block

8.	Column-I (Electronic configuration of element)	Column-II (Correct description)
	(A) $1s^1, 2s^2, 2p^2, 3s^1$	(P) s or p block element
	(B) $1s^2, 2s^2, 2p^6, 3s^2, 3p^1$	(Q) III period element
	(C) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5, 4s^1$	(R) Group number VI in the long form periodic table
	(D) $1s^2, 2s^1, 2p^3, 3s^1, 3p^3, 4s^1$	(S) d-block element
		(T) Valence shell electron ≥ 3

9. What are the blocks groups and period for the following E.C. in periodic table :

- | | |
|--|---|
| (a) $1s^2 2s^2 2p^6 3s^1 3p^3 3d^2$ | (b) $1s^2 2s^1 2p^1$ |
| (c) $1s^2$ | (d) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2$ |
| (e) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$ | (f) $[\text{Xe}] 6s^2 5d^1$ |
| (g) Actinium: $[\text{Rn}] 6d^1 7s^2$ | (h) Thorium: $[\text{Rn}] 6d^2 7s^2$ |

10. What is the value of $(n + l)$ for the unpaired e^- in an atom of an element which is present in the third period and seventeenth group of the periodic table.

Answer Key

- | | | | | | | | | | | | | |
|-----|---|----|--|----|--|----|---|----|---|----|--|--|
| 1. | B | 2. | A | 3. | D | 4. | C | 5. | B | 6. | ABC | |
| 7. | (A) S ; (B) R ; (C) P ; (D) Q | | | | | 8. | (A) PT ; (B) PQT ; (C) RST ; (D) PQ | | | | | |
| 9. | (a) block - P-block
Group - 16
Period - 3 | | (b) block - s-block
group - 2
period - 2 | | (c) block - s-block
group- 18
period - 1 | | (d) block - d-block
Group - 12
Period - 4 | | (e) block - d-block
group - 6
period -4 | | (f) block - d-block
group- 3
period -6 | |
| | (g) block - d-block
Group - 3
Period - 7 | | (h) block - f-block
group - 3
period -7 | | | | | | | | | |
| 10. | 4 | | | | | | | | | | | |

Periodic Classification-TPA-2

Only one correct :

1. For the element X, student Surbhi measured its radius as 102 nm, Mr. Gupta as 113 nm and Mr. Agarwal as 100 nm using same apparatus. Their teacher explained that measurements were correct by saying that recorded values by three students were
(A) Crystal, vander Waal and covalent radii (B) Covalent, crystal and vander Waal radii
(C) Vander Waal, ionic and covalent (D) None is correct
2. Which of the following combination contains only isoelectronic series :
(A) N^{3-} , O^{2-} , Cl^- , Ne (B) F^- , Ar, S^{2-} , Cl^-
(C) P^{3-} , S^{2-} , Cl^- , Ar (D) N^{3-} , F^- , O^{2-} , Ar
3. Which among the following species has the same number of electrons in its outermost as well as penultimate shell ?
(A) Mg^{2+} (B) O^{2-} (C) F^- (D) Ca^{2+}
4. The specie having smallest ionic radius is :-
(A) Al^{3+} (B) Ba^{2+} (C) K^+ (D) Mg^{2+}

More than one correct :

5. The correct order of radii is :
(A) $\text{N} < \text{Be} < \text{B}$ (B) $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$ (C) $\text{Na} > \text{Li} < \text{K}$ (D) $\text{Fe}^{2+} > \text{Fe}^{3+} > \text{Fe}^{4+}$
6. Choose the **correct** statement.
(A) Be and Al are not in same group.
(B) All the transition metal correspond to d-block.
(C) Be and Al are having lot of similarities in their properties.
(D) The atomic radius gradually decreases from Sc to Zn.

Subjective :

7. A monoatomic anion of unit charge contain 45 neutrons and 36 electrons. What is atomic mass number of element and in which group of periodic table does it lie.
8. Arrange in decreasing order of atomic size : Na, Cs, Mg, Si, Cl.
9. In the ionic compound KF, the K^+ and F^- ions are found to have practically radii, about 1.34 Å each. What do you predict about the relative covalent radii of K and F ?
10. Which one of the following pair would have a large size :
(i) K or K^+ (ii) Br or Br^- (iii) O^{2-} or F^- (iv) Li^+ or Na^+
(v) P or As (vi) Na^+ or Mg^{+2}

Answer Key

1. A 2. C 3. D 4. A 5. BCD 6. ABC
7. At mass no. 80, Group – 17 8. $\text{Cs} > \text{Na} > \text{Mg} > \text{Si} > \text{Cl}$
9. Radius of K > 1.34 Å, Radius of F < 1.34 Å
10. (i) K, (ii) Br^- , (iii) O^{2-} , (iv) Na^+ , (v) As, (vi) Na^+

Periodic Classification-TPA-3**Only one correct :**

1. Find the highest ratio of IP values of given pair of elements :
(A) He : Ne (B) Ne : Ar (C) He : Xe (D) Kr : Xe
2. Which one of the following electronic configuration of an atom has the highest ionisation energy.
(A) $1s^2 2s^2 2p^3$ (B) $1s^2 2s^2 2p^6 3s^1$
(C) $1s^2 2s^2 2p^6$ (D) $1s^2 2s^2 2p^5$
3. Give the correct order of initials **T** (true) or **F** (false) for following statements.
(I) Top positions of Lothar-Mayer's atomic volume curve are occupied by Alkali metals.
(II) Number of elements presents in the fifth period of the periodic table are 32.
(III) 2nd I.P. of Mg is less than the 2nd I.P. of Na.
(IV) A p-orbital can take maximum of six electrons.
(A) TFTF (B) TFFT (C) FFTF (D) TTFF
4. In crystal of which of the following ionic compound, would you expect the maximum distance between centre of cation and anion.
(A) CsF (B) CsI (C) LiI (D) LiF
5. The ionic radii of F, F^- , O^{2-} are in the order of :
(A) $O^{2-} < F^- < F$ (B) $F^- > O^{2-} > F$ (C) $O^{2-} > F^- > F$ (D) $O^{2-} = F^- > F$
6. The ionic radii of N^{3-} , O^{2-} and F^- are respectively given by :
(A) 1.36, 1.40, 1.71 (B) 1.36, 1.71, 1.40
(C) 1.71, 1.40, 1.36 (D) 1.71, 1.36, 1.40

Assertion & Reason :

7. **Statement-1** : The ionisation potential of Sn is greater than Pb.
Statement-2 : Usually ionisation energy decreases down the group.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

8. **Statement-1** : $(I.E.)_n$ of an atom is always greater than $(I.E.)_{n-1}$ (n is integer number)
Statement-2 : ne/Z ratio decreases on successive elimination of electrons.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
9. **Arrange the following periodic properties with justification.**
- (a) Increasing order of first ionization energy N, O, F
 (b) Increasing order of ionization energy Cl^- , K^+ , S^{2-} , Ca^{2+}
 (c) Increasing order of ionization energy Li^+ , Be^+ , B^+ , C^+ , N^+ , O^+ , F^+
 (d) Increasing order of ionization energy Fe, Fe^{+2} , Fe^{+3}
 (e) Increasing order of first ionization potential Mg, Al, Si, Na
10. Arrange the following in increasing order of radii
- (i) I, I^+ , I^- (ii) C, N, Si, P
 (iii) O^{2-} , N^{3-} , F^- (iv) Be^{+2} , Cl^- , S^{2-} , Na^+ , Mg^{2+}
 (v) B, Be, Li, Na (vi) Cl^- , S^{2-} , Ca^{2+}
 (vii) Mg^{2+} , Al^{3+} , Na^+ , O^{2-} , F^- (viii) S, O, Se, F

Answer Key

1. C 2. C 3. A 4. B 5. C 6. C 7. D
8. A
9. (a) $O < N < F$ (b) $Ca^{+2} > K^+ > Cl^- > S^{2-}$
 (c) $Be^+ < C^+ < B^+ < N^+ < F^+ < O^+ < Li^+$ (d) $Fe < Fe^{+2} < Fe^{+3}$
 (e) $Na < Al < Mg < Si$
10. (i) $I^+ < I < I^-$ (ii) $N < C < P < Si$
 (iii) $F^- < O^{2-} < N^{3-}$ (iv) $Be^{+2} < Mg^{+2} < Na^+ < Cl^- < S^{2-}$
 (v) $B < Be < Li < Na$ (vi) $Ca^{+2} < Cl^- < S^{2-}$
 (vii) $Al^{+3} < Mg^{+2} < Na^+ < F^- < O^{2-}$ (viii) $F < O < S < Se$

Periodic Classification-TPA-4

Only one correct :

1. Which transition involves maximum amount of energy (M is a metal)
- (A) $M^-(g) \longrightarrow M(g) + e$ (B) $M(g) \longrightarrow M^+(g) + e$
(C) $M^+(g) \longrightarrow M^{+2}(g) + e$ (D) $M^{+2}(g) \longrightarrow M^{+3}(g) + e$
2. The first five ionization energies of an element are 9.1, 16.2, 24.5, 35 and 205.7 eV respectively. Then number of valence electron in the atom is
- (A) 2 (B) 3 (C) 4 (D) 5
3. The correct order of electron affinity for the different families is
- (A) Halogen > carbon > nitrogen > oxygen
(B) Halogen > oxygen > nitrogen > carbon
(C) Halogen > nitrogen > carbon > oxygen
(D) Halogen > oxygen > carbon > nitrogen
4. $\frac{A_0}{2}$ atoms of X (g) are converted into $X^+(g)$ by energy E_1 . $A_0/2$ atoms of $X(g)$ are converted into $X^-(g)$ by energy E_2 . Hence ionisation potential and electron affinity of X (g) are
- (A) $\frac{2E_1}{A_0} \cdot \frac{2(E_1 - E_2)}{A_0}$ (B) $\frac{2E_1}{A_0} \cdot \frac{2E_2}{A_0}$
(C) $\frac{(E_1 - E_2)}{A_0} \cdot \frac{2E_2}{A_0}$ (D) None
5. First, second and third I.P. values are 10 eV, 15 eV and 150 eV. Element can be
- (A) Be (B) B (C) F (D) Na

6. Which of the following case the size ratio is minimum :
- (A) Li^- / Li (B) H^- / H
 (C) Na^- / Na (D) Can not be predicted
7. Which of the following process is associated with best possibility of the energy release.
- (A) $\text{Li} \longrightarrow \text{Li}^+ + \text{e}^-$ (B) $\text{O}^- + \text{e}^- \longrightarrow \text{O}^{2-}$ (C) $\text{Cl}^+ + \text{e}^- \longrightarrow \text{Cl}$ (D) $\text{Be} + \text{e}^- \longrightarrow \text{Be}^-$

Assertion & Reason :

8. **Statement-1:** Nitrogen atom has higher ionization energy than fluorine atom.
Statement-2: Nitrogen atom has extra stable electronic configuration due to half filled p-subshell.
- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

True & False :

9. State True or False with explanation for the following questions
- (a) E.A. of $\text{M}^{+}(\text{g})$ ion and I.E. of $\text{M}(\text{g})$ atom are equal.
 (b) EA_1 of sulphur is more than EA_1 of oxygen.
 (c) I.E. of Pb is more than Sn even though Pb is larger atom than Sn.

Subjective :

10. (a) Which has greater IE_1 ? Na^+ or Ne. Explain
 (b) Which has greater IE_2 ? O or N. Explain
 (c) Which has greater IE_2 ? Li or Be. Explain

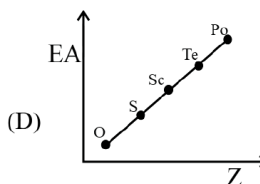
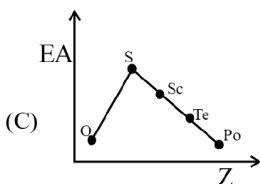
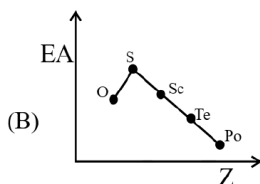
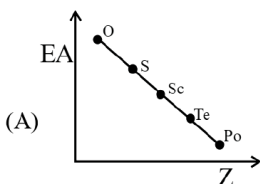
Answer Key

- | | | | | | | | | | | | | | |
|----|---|----|----------|----------|----------|-----|-------------------|-------|--------|----|---|----|---|
| 1. | D | 2. | C | 3. | D | 4. | B | 5. | A | 6. | C | 7. | C |
| 8. | D | 9. | (a) True | (b) True | (c) True | 10. | (a) Na^+ | (b) O | (c) Li | | | | |

Periodic Classification-TPA-5

Single Correct Type:

1. Select correct graph of first electron affinity values of oxygen family



Match the column

Column-I

Valence electronic configuration

- (a) ns^1
 (b) ns^2
 (c) ns^2np^1
 (d) ns^2np^2

Column-II

Successive ionization energies

- (p) 19,27,36,48,270
 (q) 16,28,34,260
 (r) 18,26,230,250
 (s) 14,200,220,240

Select the correct options:

- A) a – s , b-r, c-p, d-q
 B) a – s , b-r, c-q, d-p
 C) a – p , b-r, c-s, d-q
 D) a – s , b-q, c-p, d-r

Consider the following representation based on long form of periodic table.

		P	
S		X	Q
		R	

Here P,Q,R and S are up, right, down and left elements with respect to the central element 'X' respectively. According to above representation the incorrect match is/are:

- | | | | | |
|-------|----|----|----|----|
| X | P | Q | R | S |
| A) Ge | Si | As | Sn | Ga |
| B) Ge | Se | I | Po | Sb |
| C) Sb | As | Te | Bi | Sn |
| D) In | Ga | Sn | Tl | Cd |

4. Which of the following order is incorrect against the property indicated:
- A) $\text{Mg} < \text{Ar} < \text{Na}$ (2nd I.E.) B) $\text{Be} < \text{F} < \text{Cl}$ ($|\Delta H_{\text{eg}}|$)
- C) $\text{Rb} < \text{Na} < \text{K} > \text{Ca}$ (atomic radius) D) $\text{P} < \text{S} < \text{N}$ (electronegativity)
5. Give the correct order of initials **T** (true) or **F** (false) for following statements.
- (I) Top positions of Lothar-Mayer's atomic volume curve are occupied by Alkali metals.
- (II) Number of elements presents in the fifth period of the periodic table are 32.
- (III) 2nd I.P. of Mg is less than the 2nd I.P. of Na.
- (IV) A p-orbital can take maximum of six electrons.
- A) TFTF B) TFFT C) FFTF D) TTFF

More than one correct type

6. There are four elements 'p', 'q', 'r' and 's' having atomic numbers $Z-1$, Z , $Z+1$ and $Z+2$ respectively. If the element 'q' is an inert gas, select the correct answers from the following statements.
- A) 'p' has most negative electron gain enthalpy in the respective period.
- B) 'r' is an alkali metal
- C) 's' exists in +2 oxidation state.
- D) If "r" belongs to third period 'p' is able to show only one oxidation state.
7. Consider following ionisation steps:
- $$\text{M(g)} \rightarrow \text{M}^+(\text{g}) + \text{e}^- \quad \Delta H = 100\text{eV}$$
- $$\text{M(g)} \rightarrow \text{M}^{2+}(\text{g}) + 2\text{e}^- \quad \Delta H = 250\text{eV}$$
- Select correct statement(s):
- A) $(\text{IE})_1$ of M(g) is 100eV B) $(\text{IE})_1$ of $\text{M}^+(\text{g})$ is 150eV
- C) $(\text{IE})_2$ of M(g) is 250eV D) $(\text{IE})_2$ of M(g) is 150eV
8. Which of the following is/are correct relation?
- A) Basic character of element $\propto$ Effective nuclear charge $^{-1}$
- B) Acidic character of element $\propto$ Atomic size $^{-1}$
- C) Acidic character of element $\propto$ Size of atoms
- D) Acidic character of element $\propto$ Metallic character

Numerical Type:

13. Given

Bond energy of F – F bond = 38 Kcal/mol⁻¹

Bond energy of Cl – Cl bond = 58 Kcal/mol⁻¹

Bond energy of F – Cl bond = 61 Kcal/mol⁻¹

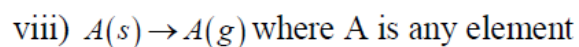
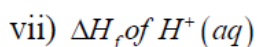
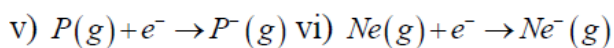
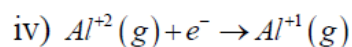
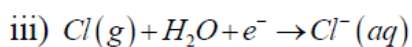
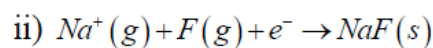
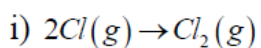
Electronegativity of fluorine = 4

Calculate the electronegativity of chlorine atom. (

$\sqrt{38} = 6.16, \sqrt{58} = 7.62, \sqrt{2} = 1.41, \sqrt{7} = 2.64$) (geometrical mean is more accurate than arithmetic mean)

14. The internuclear distance ($x \text{ \AA}$) between C and H in C – H bond is determined by using Shomaker & Stevenson equation. If internuclear distance between the two carbon atoms in C – C bond is $2.4 \text{ \AA}$ and between the two hydrogen atoms in H – H bond is $0.8 \text{ \AA}$. Given electronegativities of C and H are 2.4 and 2.1 respectively. Calculate the value of $1000x$.

15. The number of processes that are definitely exothermic (energy is released) out of the following processes are



1. (C) 2. (B) 3. (B) 4. (C) 5. (A) 6. (ABCD) 7. (ABD) 8. (AB) 9. (ABCD) 10. (AD)
11. (A) 12. (A) 13. (3.2 t 3.32) 14. (1573) 15. (5)